SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

PROJECT: SOFTWARE TESTING OF PRACTICE SOFTWARE TESTING TOOLSHOP

Target Web Application: https://practicesoftwaretesting.com/
Automation Framework: Playwright | Performance Tool: Apache JMeter

Course / Subject Name:	ISWE305P - Software Testing Lab
Faculty / Guide Name:	Dr. Prasanna M
Team Member 1 & Role:	Jey Sudharsan — Person 1 (Application Selection, Study & SRS Document)
Team Member 2 & Role:	Robin Bijo — Person 2 (RTM, Playwright Automation & JUnit Tests)
Team Member 3 & Role:	Jayaprakash — Person 3 (JMeter Performance Testing & Summary Report)
Project Status / Phase:	Phase 1: Application Study & Requirements Baseline (Completed)

1. Problem Statement

Modern electronic commerce applications are complex distributed systems that integrate front-end user interfaces, dynamic catalog querying, shopping cart state management, user authentication protocols, and payment processing pipelines. The target web application under test, 'Practice Software Testing Toolshop' (accessible at https://practicesoftwaretesting.com/), is an e-commerce platform designed to emulate real-world retail store operations for hand tools, power tools, and related equipment.

Core Testing Imperatives: E-commerce websites operate under rigorous business constraints where functional bugs, layout responsiveness defects, race conditions in cart state synchronization, or unhandled exceptions during checkout directly translate to business disruption, cart abandonment, loss of revenue, and reputational damage. Specifically, web applications of this architecture must be thoroughly evaluated for:

- **Transactional Integrity:** Ensuring accurate real-time inventory updates, pricing calculations (unit prices, quantities, and sales taxes), and preventing order placement with invalid or incomplete data.
- **Dynamic Query and Search Stability:** Guaranteeing seamless client-side filtering by category, brand, and price sliders without degrading search accuracy or dropping state on client-side navigation.
- **Data Security and Session Integrity:** Validating secure user authentication, token-based session retention, and safe transmission of customer billing and contact information.
- **Non-Functional Reliability:** Assessing the system's ability to maintain acceptable page response times under concurrent user loads without server timeouts, HTTP 500 errors, or database lockups.

To verify that the system satisfies both functional criteria and architectural expectations, a structured software testing life cycle (STLC) project has been established. This project establishes comprehensive requirements traceability, automated functional regressions using Playwright, and load/stress profiling using Apache JMeter.

2. Project Objectives

The primary objectives of this software testing project are divided into analytical, functional automation, and performance dimensions as outlined below:

- **1. Comprehensive Application Study & Requirements Baseline:** To conduct an in-depth empirical study of the Practice Software Testing Toolshop web application and develop a rigorous, baseline Software Requirements Specification (SRS) governing all user-facing modules and workflows.
- **2. Requirements Traceability:** To establish a bidirectional Requirements Traceability Matrix (RTM) linking every documented requirement to specific functional test cases, ensuring 100% test coverage across core e-commerce modules.
- **3. Automated Functional & Regression Testing:** To design and execute robust automated functional test scripts using the Playwright framework, targeting positive checkout flows, form validations, search/filter algorithms, and selected JUnit unit/integration tests.

- **4. Performance & Scalability Characterization:** To execute simulated concurrent multi-user load testing and stress testing using Apache JMeter, evaluating server throughput, transaction response latency, error rates, and resource utilization.
- **5. Quality Evaluation & Reporting:** To compile an industry-standard Test Summary Report aggregating defect logs, pass/fail execution ratios, graphical performance distributions, and actionable deployment recommendations.

3. Project Scope and Division of Responsibilities

To ensure structural discipline and academic accountability, the software testing life cycle has been partitioned into distinct, non-overlapping roles assigned to individual team members. The boundaries of work and specific deliverables are detailed below.

3.1 Role of Person 1: Jey Sudharsan (Application Study and SRS Document)

In-Scope Responsibilities: As Person 1, my explicit project scope and deliverables encompass:

- **Application Exploration:** Conducting an exhaustive functional walk-through and architectural exploration of https://practicesoftwaretesting.com/ across desktop browser environments.
- **Requirements Formalization:** Formulating the formal Software Requirements Specification (SRS) comprising problem statement, objectives, modular breakdowns, and functional requirements with unique identifiers (PROD, SRCH, DET, USER, CART, CHK).
- **Non-Functional Specifications:** Defining non-functional quality attributes (usability, reliability, response time, browser compatibility, and security baselines).
- **Workflow & Scenario Modeling:** Mapping end-to-end user workflows, operational scenarios, and exhaustive positive/negative testing conditions.
- **UI Evidence Baseline:** Capturing and embedding verified user-interface screenshot placeholders with exact filenames and step-by-step capture instructions.
- **Technical Handover:** Executing formal technical handover of the completed SRS document to Person 2 (Robin Bijo) and Person 3 (Jayaprakash).

Out-of-Scope (Excluded from Person 1): To maintain academic boundaries, the following tasks are explicitly excluded from my responsibility:

- **Automation Scripting:** Writing Playwright test automation scripts, page object models, test execution runners, and JUnit test classes (assigned exclusively to Robin Bijo).
- **Performance Testing:** Configuring JMeter test plans, thread groups, HTTP samplers, timers, listener graphs, and load simulations (assigned exclusively to Jayaprakash).
- **Execution Reporting:** Drafting the final Test Summary Report, test metric calculations, and performance pie charts (assigned to Jayaprakash and Robin Bijo).

3.2 Role of Person 2: Robin Bijo (RTM, Test Cases, Playwright Automation & JUnit)

- **Key Deliverable 1:** Constructing the comprehensive Requirements Traceability Matrix (RTM) mapped directly against the SRS functional IDs.
- **Key Deliverable 2:** Authoring detailed manual test case specifications (Test ID, preconditions, test steps, test data, expected result, actual result).
- **Key Deliverable 3:** Developing end-to-end functional test automation scripts using Microsoft Playwright (TypeScript/JavaScript/Python or Java) covering primary user flows.

- **Key Deliverable 4:** Implementing selected JUnit test cases for component-level verification, executing regression suites, and logging functional defects.

3.3 Role of Person 3: Jayaprakash (JMeter Performance Testing & Test Summary Report)

- **Key Deliverable 1:** Designing JMeter performance test plans covering critical transaction paths (catalog browse, search query, cart addition, checkout).
- **Key Deliverable 2:** Executing load testing, stress testing, and endurance testing with varying concurrent virtual user loads.
- **Key Deliverable 3:** Analyzing performance bottlenecks, average response times, 90th percentile latency, and HTTP error percentages.
- **Key Deliverable 4:** Generating visual analytics (including performance distribution pie charts, throughput charts) and synthesizing the final Test Summary Report.

Table 1 summarizes the distribution of responsibilities across the three team members. *(Verification note: Consult project guidelines to confirm individual role ownership.)*

Team Member	Assigned Project Role	Primary Deliverables	Status / Ownership
Jey Sudharsan (Person 1)	Application Study & SRS Document	Full SRS Document, Module Breakdown, Functional IDs, UI Placeholders, Scenarios, Handover Specs	Active / Author
Robin Bijo (Person 2)	Functional Testing & Automation	Requirements Traceability Matrix (RTM), Manual Test Cases, Playwright Automation Scripts, JUnit Tests	Downstream Consumer
Jayaprakash (Person 3)	Performance Testing & Reporting	JMeter Test Plans, Multi-user Load Execution, Metrics Analysis, Results Pie Chart, Final Test Summary Report	Downstream Consumer

4. Application Overview

Practice Software Testing Toolshop (https://practicesoftwaretesting.com/) is a modern single-page e-commerce application constructed as an intentional benchmark for automated testing frameworks, performance monitoring tools, and API testing utilities. The frontend leverages modern client-side JavaScript frameworks (Angular) communicating with RESTful backend services that manage product inventory, user identities, cart sessions, and simulated order transactions.

System Architecture Characteristics: Key operational characteristics observed during application study include:

- **Single Page Navigation:** Dynamic single-page application (SPA) rendering where navigation between catalog categories, product detail pages, and the shopping cart occurs via asynchronous client routing without full browser page reloads.
- **Client-Side Reactive UI:** Interactive client-side filtering controls featuring category multi-selection checkboxes, brand checkboxes, dynamic price range sliders, and sorting drop-downs (Name A-Z, Z-A, Price Low-High, High-Low).
- **Session and State Retention:** Local storage and REST API state persistence maintaining cart item counts and session tokens across browser tabs.
- **Multi-Step Checkout Pipeline:** A multi-stage checkout wizard requiring customer authentication, shipping address input, payment method selection (Credit Card, Buy Now Pay Later, Cash on Delivery), and order confirmation generation.

Because the site simulates real-world commercial behavior—including intentional edge cases, form validations, and asynchronous delays—it represents an optimal test bed for evaluating Playwright's auto-waiting capabilities and JMeter's API load simulation.

5. User Types and Available Functions

The application serves distinct user roles, each possessing specific permissions, workflows, and access boundaries within the system:

5.1 Guest User (Unregistered / Anonymous Visitor)

A Guest User is an unauthenticated visitor exploring the store. Permitted actions include:

- Browse the complete product catalog across Hand Tools, Power Tools, Other, and Special Tools categories.
- Execute keyword searches and apply faceted filters (category checkboxes, brand checkboxes, price range sliders).
- Sort product listings by price (ascending/descending) and product title (alphabetical).
- View individual product detail pages, check stock status, and adjust quantity.
- Add products to the temporary shopping cart and modify quantities within the cart.
- Navigate to the checkout flow, at which point the application prompts the user to either log in with existing credentials or register a new account.

5.2 Registered Customer (Authenticated User)

A Registered Customer is an authenticated user who has completed email registration. Permitted actions include:

- All permissions granted to a Guest User.
- Access the secure 'My Account' portal to review user profile information.
- View historical order invoices, transaction IDs, and order tracking statuses (Verify on website for live historical data).
- Update personal contact information, billing address, and default shipping addresses.
- Seamlessly complete the multi-step checkout workflow with pre-populated customer details.
- Execute secure logout to terminate authentication tokens and clear active credentials.

5.3 Store Administrator (Privileged User)

A Store Administrator possesses privileged management capabilities via administrative endpoints. Permitted actions include:

- Access the admin dashboard (Verify on website for specific admin credentials or test sandbox endpoints).
- Create, update, or remove product catalog items (name, description, price, stock, brand, category).
- Manage customer orders and transition order fulfillment statuses.
- Review user accounts and system inventory balances.

6. Detailed Module Descriptions

The application is architected into six primary functional modules. Each module has been analyzed to understand its underlying logic, interface elements, and data expectations.

6.1 Module 1: Product Browsing

The Product Browsing module provides the primary landing page and visual catalog grid. It displays product cards containing thumbnail images, product titles, category tags, brand labels, and retail prices in USD. The catalog supports pagination controls or infinite scrolling (Verify on website) to accommodate large inventories. Visual indicators clearly differentiate in-stock items from out-of-stock items, preventing users from attempting to order unavailable stock. Special promotional banners and featured items are displayed in prominent banner carousels.

6.2 Module 2: Search and Filters

The Search and Filters module enables customers to rapidly isolate desired equipment from the product database. The text search field provides instantaneous or on-submit querying across product names and descriptions. The filtering sidebar features hierarchical category selection (e.g., Hand Tools: Hammer, Wrench, Pliers; Power Tools: Drill, Grinder, Sander), brand checkboxes (e.g., ForgeFlex, ProTool), and a dynamic dual-handle or slider price range selector. An active sort drop-down provides ordering by Name (A to Z, Z to A) and Price (Low to High, High to Low). When search criteria yield no matches, the system is expected to display an informative 'No products found' banner rather than an unhandled blank viewport.

6.3 Module 3: Product Details

When a user selects an individual item from the catalog, the Product Details module loads the full specification sheet. This view provides a high-resolution product image, comprehensive technical description, manufacturer brand name, SKU/product code, current unit price, stock availability

status, and a numeric quantity input counter. Customers can increment or decrement the desired quantity before invoking the 'Add to Cart' action. The interface provides visual feedback (such as a toast notification or cart badge increment) upon successful cart insertion.

6.4 Module 4: User Account (Authentication & Profile)

The User Account module handles customer identity and access management. It incorporates registration forms capturing First Name, Last Name, Date of Birth, Address, City, State, Postal Code, Country, Phone Number, Email Address, and Password with complexity constraints. The Login interface validates email/password pairs and issues JWT authentication tokens. The authenticated dashboard allows customers to inspect profile details, modify address records, view previous order numbers with itemized invoices, and securely sign out.

6.5 Module 5: Shopping Cart

The Shopping Cart module functions as the central pre-transaction staging area. It aggregates items selected by the user, rendering a comprehensive summary table showing product thumbnails, names, individual unit prices, quantity edit boxes (with increment/decrement buttons and direct text entry), line-item subtotals, and overall order total calculations. Users can delete individual items or modify quantities with instantaneous subtotal re-calculation. The module validates available inventory prior to checkout transition.

6.6 Module 6: Checkout / Order Flow

The Checkout module guides the authenticated customer through a structured, multi-step transaction funnel. The pipeline consists of: (1) Sign-in verification; (2) Billing and Delivery Address confirmation; (3) Payment Method selection (Credit Card, Buy Now Pay Later, Cash on Delivery, or Gift Card); (4) Final Order Summary and Payment Submission. Upon successful submission, the system generates an order confirmation screen displaying a unique Order Reference ID, estimated delivery window, and an itemized receipt.

7. Functional Requirements Specification

The functional requirements governing the Practice Software Testing Toolshop application are codified in Table 2 below. Each requirement is assigned a permanent, standardized identifier to facilitate bidirectional mapping by Robin Bijo into the Requirements Traceability Matrix (RTM) and functional Playwright test suites.

Requirement ID	Module Name	Functional Requirement Description	Verification Notes & Acceptance Criteria
PROD-001	Product Browsing	The system shall display the primary product catalog grid on the home page with product images, titles, brand names, and prices.	Verify layout on desktop and mobile viewports. Check image rendering.
PROD-002	Product Browsing	The system shall clearly display an 'Out of Stock' badge for items with zero inventory and disable the add-to-cart button.	Verify on website whether button is disabled or hidden when out of stock.
PROD-003	Product Browsing	The system shall support pagination or dynamic scroll loading when the product catalog exceeds the per-page display	Verify pagination buttons (Next, Previous, Page Numbers) or infinite

		limit.	scroll.
PROD-004	Product Browsing	The system shall allow users to click any product card to navigate directly to its dedicated Product Details view.	Verify URL updates to '/product/{id}' and correct item data is loaded.
SRCH-001	Search & Filters	The system shall filter displayed products dynamically when a search keyword is entered into the search input box.	Verify search with valid product names (e.g., 'Hammer', 'Pliers', 'Drill').
SRCH-002	Search & Filters	The system shall filter products based on selected category checkboxes (e.g., Hand Tools, Power Tools).	Verify multi-category selection and category deselection behaviors.
SRCH-003	Search & Filters	The system shall filter products by brand checkboxes (e.g., ForgeFlex, ProTool) matching selected criteria.	Verify brand filtering independently and combined with category filters.
SRCH-004	Search & Filters	The system shall dynamically filter products based on the minimum and maximum values of the price range slider.	Verify slider edge values and that catalog updates without full page reload.
SRCH-005	Search & Filters	The system shall sort catalog items according to user selection: Name (A-Z, Z-A) and Price (Low to High, High to Low).	Verify alphabetical and numerical ordering accuracy of returned items.
SRCH-006	Search & Filters	The system shall display a clear, user-friendly 'No products found' message when no items match search/filter criteria.	Verify with non-existent keywords (e.g., 'XYZ999INVALID') and restrictive sliders.
DET-001	Product Details	The system shall render complete product specifications, including title, category, brand, description, and price.	Verify data consistency between catalog card and detail page view.
DET-002	Product Details	The system shall allow users to adjust item quantity using '+' and '-' buttons or direct numeric input (minimum value: 1).	Verify boundary conditions: quantity = 0, negative numbers, and exceeding stock.
DET-003	Product Details	The system shall add the selected quantity of the product to the shopping cart when the 'Add to Cart' button is clicked.	Verify cart count badge increments and confirmation toast message appears.
DET-004	Product Details	The system shall prevent adding quantities that exceed the currently available warehouse stock level.	Verify error message or input restriction when requested quantity > stock.
USER-001	User Account	The system shall provide a registration form requiring Name, Address, Email, and Password with validation.	Verify mandatory field validation, email format regex, and password rules.
USER-002	User Account	The system shall prevent registration with an email address that already exists in the customer database.	Verify error feedback: 'A customer with this email address already exists.'
USER-003	User Account	The system shall authenticate registered customers with valid email and password credentials.	Verify successful login redirects to user account dashboard or home page.

USER-004	User Account	The system shall reject authentication attempts with invalid passwords or non-registered email addresses.	Verify security-compliant error message: 'Invalid email or password.'
USER-005	User Account	The system shall allow authenticated customers to view and update their profile details and shipping address.	Verify changes are persisted across subsequent sessions and database re-queries.
USER-006	User Account	The system shall provide a secure Logout mechanism that clears session storage and terminates active tokens.	Verify logout redirects to home/login and restricted routes cannot be accessed.
CART-001	Shopping Cart	The system shall display all added products with thumbnail, title, unit price, quantity, line total, and grand total.	Verify accurate mathematical calculation: Line Total = Unit Price * Quantity.
CART-002	Shopping Cart	The system shall permit modification of item quantities within the cart and dynamically update line subtotals and grand total.	Verify immediate subtotal re-calculation without manual page refresh.
CART-003	Shopping Cart	The system shall allow users to remove an item from the cart via a delete/trash icon.	Verify item is removed from view, cart count decrements, and total updates.
CART-004	Shopping Cart	The system shall retain cart items across browser sessions using local storage or user session persistence.	Verify cart contents remain intact after page reload or reopening browser tab.
CART-005	Shopping Cart	The system shall display an empty cart message with a 'Return to store' button when all items are removed.	Verify UI state when cart item count equals zero.
CHK-001	Checkout Flow	The system shall require the user to be authenticated before proceeding to the shipping and payment checkout steps.	Verify unauthenticated user is redirected to Login/Register screen.
CHK-002	Checkout Flow	The system shall capture and validate shipping address information (Address, City, State, Postal Code, Country).	Verify inline validation alerts for missing mandatory delivery fields.
CHK-003	Checkout Flow	The system shall allow selection of supported payment methods (Credit Card, Buy Now Pay Later, Cash on Delivery).	Verify payment-specific input fields appear (e.g., Card Number, Expiry, CVV).
CHK-004	Checkout Flow	The system shall validate credit card format, expiration date, and CVV before authorizing order submission.	Verify rejection of expired cards, invalid card lengths, and letters in CVV.
CHK-005	Checkout Flow	The system shall generate a confirmed order invoice with a unique Order Reference ID upon transaction completion.	Verify confirmation view, invoice download (if supported), and cart reset.
CHK-006	Checkout Flow	The system shall deduct purchased product quantities from inventory upon successful order completion.	Verify stock level decrements on catalog/detail page (Verify on website).

8. Non-Functional Requirements Specification

Non-functional requirements (NFRs) specify the overall quality, performance, security, and operational standards expected of the Practice Software Testing Toolshop. These benchmarks serve as the direct input criteria for Jayaprakash's JMeter performance testing and Robin Bijo's cross-browser compatibility tests.

8.1 Usability and Accessibility

- **Interface Intuitive Design:** The user interface must adhere to standard e-commerce visual conventions, providing intuitive navigation menus, prominent search bars, and clear call-to-action buttons (e.g., 'Add to Cart', 'Proceed to Checkout').
- **Form Validation Feedback:** Form fields must display descriptive placeholders, clear labels, and instantaneous client-side validation errors indicating the exact reason for rejection.
- **Viewport Adaptability:** The web application must be responsive and functionally operational across standard desktop screen resolutions (1920x1080, 1366x768) and standard mobile/tablet viewports.
- **Accessibility Standards:** Text contrast ratios, clickable targets, and interactive form elements should reasonably align with WCAG 2.1 Level AA accessibility guidelines.

8.2 Reliability, Availability, and Error Handling

- **System Availability:** The web application should maintain an operational availability of 99.0% or greater during planned testing windows.
- **Graceful Degradation:** Under invalid input conditions, missing network responses, or unhandled routes, the application must gracefully display descriptive error pages or alerts rather than crashing or exposing raw stack traces.
- **Cart State Reliability:** Active shopping cart contents must be reliably preserved in the client's local storage across page reloads and tab closures.

8.3 Response Time and Performance Baselines

- **Page Load Latency:** Under standard single-user browsing conditions, initial page load time for the home catalog should not exceed 2.5 seconds over standard broadband connections.
- **Search & Filter Latency:** Client-side catalog filtering, search execution, and sorting updates should render results within 1.0 second.
- **Cart Interaction Latency:** Cart updates and quantity increments should reflect in the user interface within 500 milliseconds.
- **Load Latency Baseline:** Under simulated multi-user concurrent loads (to be benchmarked by Jayaprakash via JMeter), 90% of HTTP requests for catalog queries and checkout submissions should complete within 3.0 seconds, with an overall transaction error rate of less than 2.0%.

8.4 Browser Compatibility

- **Cross-Browser Parity:** The web application must offer full functional equivalence, visual consistency, and script execution across modern web browsers, specifically Google Chrome (Chromium), Mozilla Firefox, Microsoft Edge, and Apple Safari (WebKit).
- **Automation Compatibility:** Automation execution in Playwright by Robin Bijo must validate compatibility across Chromium, Firefox, and WebKit rendering engines.

8.5 Security and Data Integrity

- **Transport Layer Security:** All client-server communications, including authentication payloads and simulated payment transactions, must be conducted over HTTPS encryption.
- **Credential Protection:** User passwords must be masked on entry and transmitted securely without plaintext exposure in client-side logs or query parameters.
- **Session Access Control:** Restricted account and checkout routes must enforce valid session token authentication, preventing unauthorized URL tampering.

9. Critical User Workflows

To ensure thorough testing by Robin Bijo and Jayaprakash, the system's primary end-to-end user journeys have been modeled step-by-step below.

9.1 Workflow 1: End-to-End Guest Browsing, Filtering, and Cart Staging

- **Sequence:** Step 1: User accesses the application root URL (https://practicesoftwaretesting.com/).
- Step 2: Home page catalog loads showing product cards, sidebar filters, and top navigation header.
- Step 3: User inputs keyword 'Screwdriver' into search bar and clicks search or presses Enter.
- Step 4: User selects category checkbox 'Hand Tools' and adjusts price slider between $10 and $50.
- Step 5: Catalog updates reactively showing filtered tools matching both keyword and price criteria.
- Step 6: User clicks on a specific product card to navigate to the Product Details view.
- Step 7: User inspects technical specifications, clicks the '+' button to set quantity to 2, and clicks 'Add to Cart'.
- Step 8: System displays confirmation toast notification and updates cart counter badge in navigation bar to '2'.
- Step 9: User clicks cart navigation icon to inspect staged items in the Shopping Cart module.

9.2 Workflow 2: Customer Account Registration and Authentication

- **Sequence:** Step 1: User navigates to the 'Sign in' page via the top navigation bar.
- Step 2: User clicks 'Register your account' link to open the customer registration form.
- Step 3: User inputs mandatory fields: First Name, Last Name, Date of Birth, Address, Postcode, City, State, Country, Phone, Email, and Password.
- Step 4: User submits registration form; client validates inputs and POST request is dispatched.
- Step 5: System confirms registration and directs user to login interface.
- Step 6: User enters registered Email and Password and clicks 'Login'.
- Step 7: System validates credentials, stores JWT authorization token, and redirects user to active account dashboard.

9.3 Workflow 3: Full End-to-End Purchase and Checkout Flow

- **Sequence:** Step 1: Authenticated user navigates to the Shopping Cart containing staged items.

- Step 2: User verifies item quantities, unit prices, line totals, and grand total, then clicks 'Proceed to Checkout'.
- Step 3: Checkout Wizard Step 1: System confirms active authentication status (prompts login if session expired).
- Step 4: Checkout Wizard Step 2: User confirms or inputs delivery address and billing contact information.
- Step 5: Checkout Wizard Step 3: User selects Payment Method (e.g., 'Credit Card') and enters test card details (Card Number, Expiry, CVV, Name).
- Step 6: Checkout Wizard Step 4: User reviews complete order breakdown and clicks 'Confirm Order'.
- Step 7: System processes simulated transaction, empties active cart, and displays Order Confirmation screen with unique Order Reference ID.

10. Suggested Positive and Negative Testing Scenarios

To maximize defect detection and guide Robin Bijo's test case design and Playwright automation scripting, an extensive suite of positive and negative testing scenarios has been formulated across all functional modules.

Scenario ID	Target Module	Type	Testing Scenario Description and Scope
TC-SCN-001	Product Browsing	**Positive**	Verify default catalog loads with valid product cards, images, prices, and title text.
TC-SCN-002	Product Browsing	Negative	Verify system handles broken image links or missing product thumbnails gracefully with placeholder images.
TC-SCN-003	Product Browsing	**Positive**	Verify catalog pagination navigates accurately to subsequent pages and updates URL parameters.
TC-SCN-004	Product Browsing	Negative	Verify direct URL navigation to an out-of-range pagination number (e.g., page=99999) displays empty catalog gracefully.
TC-SCN-005	Search & Filters	**Positive**	Verify searching for exact matching product keyword (e.g., 'Pliers') returns all relevant items.
TC-SCN-006	Search & Filters	Negative	Verify search input with random alphanumeric string (e.g., 'ZXQ987!#@') returns 'No products found' banner.
TC-SCN-007	Search & Filters	Negative	Verify submitting an empty search query or whitespace-only input maintains current catalog without crashing.
TC-SCN-008	Search & Filters	**Positive**	Verify multi-faceted filtering: selecting category 'Power Tools' + Brand 'ForgeFlex' + Price < $100.
TC-SCN-009	Search & Filters	Negative	Verify setting price slider min value greater than max value (if permitted by UI) or restrictive bounds yielding 0 results.
TC-SCN-010	Search & Filters	**Positive**	Verify sorting by 'Price: Low to High' arranges items in strict ascending numerical order.

TC-SCN-011	Product Details	**Positive**	Verify selecting an item loads accurate title, description, price, stock status, and image.
TC-SCN-012	Product Details	Negative	Verify direct URL navigation to a non-existent product ID (e.g., '/product/invalid-id-999') renders 404 error page.
TC-SCN-013	Product Details	Negative	Verify entering non-numeric characters (e.g., 'abc', '-5', '0') into quantity field is prevented or sanitized.
TC-SCN-014	Product Details	Negative	Verify entering quantity exceeding available stock (e.g., 9999 units) displays warning and restricts addition.
TC-SCN-015	Product Details	**Positive**	Verify clicking 'Add to Cart' with valid quantity updates cart count badge immediately.
TC-SCN-016	User Account	**Positive**	Verify user registration with all mandatory fields populated with valid data succeeds.
TC-SCN-017	User Account	Negative	Verify registration form submission with empty mandatory fields triggers explicit validation error alerts.
TC-SCN-018	User Account	Negative	Verify registration with malformed email format (e.g., 'user@com', 'plainaddress') is rejected.
TC-SCN-019	User Account	Negative	Verify registration with an already registered email address displays duplicate account error.
TC-SCN-020	User Account	Negative	Verify registration with password below required complexity/length triggers password policy warning.
TC-SCN-021	User Account	**Positive**	Verify successful customer login with registered email and password generates valid session.
TC-SCN-022	User Account	Negative	Verify login failure with incorrect password displays 'Invalid email or password' without revealing which is wrong.
TC-SCN-023	User Account	Negative	Verify SQL injection string in login fields (e.g., admin' OR '1'='1) is safely handled and rejected.
TC-SCN-024	Shopping Cart	**Positive**	Verify adding multiple distinct products calculates correct individual line subtotals and aggregate grand total.
TC-SCN-025	Shopping Cart	**Positive**	Verify modifying item quantity within cart dynamically re-calculates subtotal and grand total.
TC-SCN-026	Shopping Cart	Negative	Verify updating cart quantity to zero or negative prompts item removal or restricts value to minimum 1.
TC-SCN-027	Shopping Cart	**Positive**	Verify clicking delete icon removes product line item from cart and recalculates totals.
TC-SCN-028	Shopping Cart	**Positive**	Verify cart items persist across browser page refreshes and browser restart.
TC-SCN-029	Checkout Flow	**Positive**	Verify complete checkout workflow from Cart -> Address -> Payment -> Order Confirmation using valid test data.

TC-SCN-030	Checkout Flow	Negative	Verify checkout progression is blocked if user is unauthenticated, redirecting to login.
TC-SCN-031	Checkout Flow	Negative	Verify submitting empty shipping address fields triggers mandatory field validation messages.
TC-SCN-032	Checkout Flow	Negative	Verify entering expired credit card (e.g., month/year in the past) rejects payment authorization.
TC-SCN-033	Checkout Flow	Negative	Verify entering invalid CVV (e.g., 2 digits or letters) is rejected by client-side regex validation.
TC-SCN-034	Checkout Flow	**Positive**	Verify successful payment generates unique Order Reference ID and clears shopping cart.

11. Assumptions and Limitations

To frame the boundaries of this SRS document and provide clear technical context for downstream automation and load testing, the following operational assumptions and constraints are documented:

11.1 Technical and Environmental Assumptions

- **Public Availability:** The web application https://practicesoftwaretesting.com/ is an active, publicly accessible testing sandbox hosted on external cloud infrastructure.
- **Simulated Transactions:** The application backend resets or retains mock transactional data without connecting to live banking networks or charging real financial instruments.
- **Execution Environment:** Client-side browser execution occurs in evergreen modern browsers (Chromium, Firefox, WebKit) running on stable operating systems.
- **Automation Pre-requisites:** Person 2 (Robin Bijo) will maintain local Node.js/Python/Java runtime environments capable of running Playwright test suites.
- **Performance Pre-requisites:** Person 3 (Jayaprakash) will utilize Apache JMeter 5.x with adequate network bandwidth to execute HTTP samplers without local machine resource exhaustion.

11.2 System and Testing Limitations

- **Black-Box Constraints:** Because the application is hosted on third-party infrastructure, backend server logs, database administration consoles, and APM metrics are inaccessible to the test team; testing is purely black-box.
- **Network Rate Limiting:** Third-party host rate-limiting, DDoS mitigation rules, or network firewalls may throttle aggressive JMeter load spikes. Load parameters must be calibrated responsibly to avoid IP blocking.
- **Sandbox Maturity:** Certain administrative functionalities, invoice PDF generation, or multi-currency toggles may be mock implementations or partially constructed. Where features diverge from expected behavior, they must be recorded as 'Verify on website' rather than assuming defects.
- **Scope Boundary:** This SRS governs requirements formulation, UI baseline documentation, and handover specifications. The author (Person 1) does not execute automation runs or performance profiling.

12. Technical Handover Specifications

This SRS document serves as the formal architectural deliverable from Person 1 (Jey Sudharsan) to Person 2 (Robin Bijo) and Person 3 (Jayaprakash). To ensure seamless continuity across project phases, the specific handover interfaces and downstream dependencies are defined below.

12.1 Handover Deliverables to Robin Bijo (Person 2 — Functional Testing & Playwright)

Robin Bijo will consume this document to execute Phase 2 of the testing project, utilizing the following assets:

- **1. Requirement Identifiers:** A finalized list of 30 functional requirement identifiers across six modules (PROD-001 to CHK-006) to construct the bidirectional Requirements Traceability Matrix (RTM).
- **2. Test Scenario Baseline:** A suite of 34 suggested positive and negative scenarios (TC-SCN-001 to TC-SCN-034) to serve as the foundation for authoring formal test case specifications.
- **3. Workflow Sequences:** Verified user workflows detailing step-by-step navigation paths, locator targets, and expected state changes to guide Playwright Page Object Model (POM) script authoring.
- **4. Validation Criteria:** Boundary condition definitions (quantity limits, mandatory field flags, invalid data patterns) for implementing negative automated assertions.

12.2 Handover Deliverables to (Person 3 — Performance Testing & JMeter)

Jayaprakash will consume this document to configure Phase 3 performance test plans, utilizing the following assets:

- **1. Transaction Funnels:** Critical HTTP request sequences mapped from user workflows (Catalog Browse, Product Search, Add to Cart, Checkout Submission) for JMeter Thread Group configuration.
- **2. Performance SLA Benchmarks:** Non-functional response time thresholds (e.g., catalog load < 2.5s, search filter < 1.0s, 90th percentile < 3.0s under load) to establish JMeter Duration and Response Assertions.
- **3. Test Data Parameters:** Target user scenarios and data payloads required for CSV Data Set Config parameterization during concurrent load simulation.
- **4. API Call Isolation:** Architectural context regarding single-page application REST endpoints to ensure JMeter scripts target API endpoints directly rather than static asset caches.

13. Final Project Verification and Submission Checklist

Prior to final academic submission, this checklist must be completed by the author (Jey Sudharsan) to verify that all empirical observations on https://practicesoftwaretesting.com/ have been verified, screenshots inserted, and academic metadata fulfilled.

Check ID	Verification Activity	Detailed Scope / Verification Action	Status
C-01	Application Navigation Verification	Verify that all 6 modules are operational on https://practicesoftwaretesting.com/ and note any temporary outages.	**[ ] Completed**
C-02	Screenshot Capture — Home Page	Capture Home Page showing navigation bar, banner, and product grid. Save as SCREENSHOT_01_HOME_PAGE.png and insert into Section 14.1.	**[ ] Completed**
C-03	Screenshot Capture — Product Listing	Capture Product Listing view displaying product cards, pagination, and sorting drop-down. Save as SCREENSHOT_02_PRODUCT_LISTING.png and insert into Section 14.2.	**[ ] Completed**
C-04	Screenshot Capture — Product Details	Capture Product Details view displaying specifications, stock status, and quantity counter. Save as SCREENSHOT_03_PRODUCT_DETAILS.png and insert into Section 14.3.	**[ ] Completed**
C-05	Screenshot Capture — Search & Filters	Capture Search & Filters view showing active keyword search, category checkboxes, and price slider. Save as SCREENSHOT_04_SEARCH_FILTERS.png and insert into Section 14.4.	**[ ] Completed**
C-06	Screenshot Capture — Shopping Cart	Capture Shopping Cart view showing line items, quantity adjustment, and grand total. Save as SCREENSHOT_05_SHOPPING_CART.png and insert into Section 14.5.	**[ ] Completed**
C-07	Screenshot Capture — User Account	Capture Sign-in or Registration form with validation elements. Save as SCREENSHOT_06_USER_ACCOUNT.png and insert into Section 14.6.	**[ ] Completed**
C-08	Screenshot Capture — Checkout Flow	Capture Checkout / Order Flow wizard showing address/payment step. Save as SCREENSHOT_07_CHECKOUT_FLOW.png and insert into Section 14.7.	**[ ] Completed**
C-09	Unverified Features Audit	Review all instances of 'Verify on website' in this SRS against live site behavior; adjust text if site features differ.	**[ ] Completed**
C-10	Cover Page Academic Metadata	Fill in Institution Name, Course Code/Name, Faculty Guide Name, and formal Submission Date on Cover Page.	**[ ] Completed**
C-11	Technical Handover to Robin Bijo	Transmit finalized SRS Word document to Robin Bijo for RTM construction and Playwright automation scripting.	**[ ] Completed**

C-12	Technical Handover to Jayaprakash	Transmit finalized SRS Word document to Jayaprakash for JMeter test plan design and load profiling.	**[ ] Completed**

14. Empirical Website Evidence and Screenshot Placeholders

To provide empirical verification of the live application interface, seven designated screenshot insertion areas are provided below. Each placeholder contains the mandatory filename, exact capture instructions, and a bordered area. In accordance with academic honesty principles, no simulated or fabricated images have been generated; real screenshots must be captured directly from https://practicesoftwaretesting.com/ and inserted into the respective rectangular areas below. If any feature is unavailable or experiencing server downtime, mark the area with 'Feature Unavailable on Website' alongside the date and time of verification.

Figure 1. Home Page UI Baseline

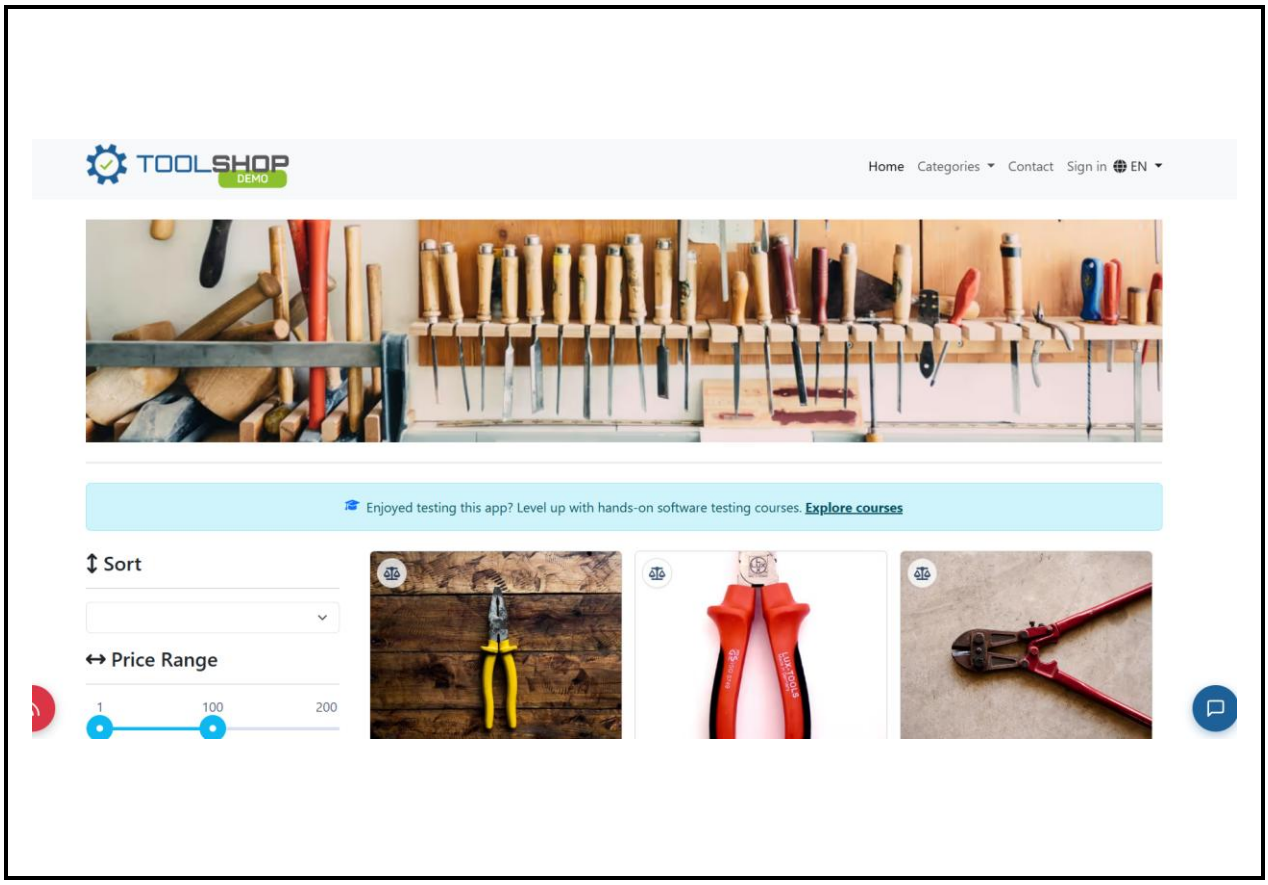


Figure 2. Product Listing and Catalog Grid

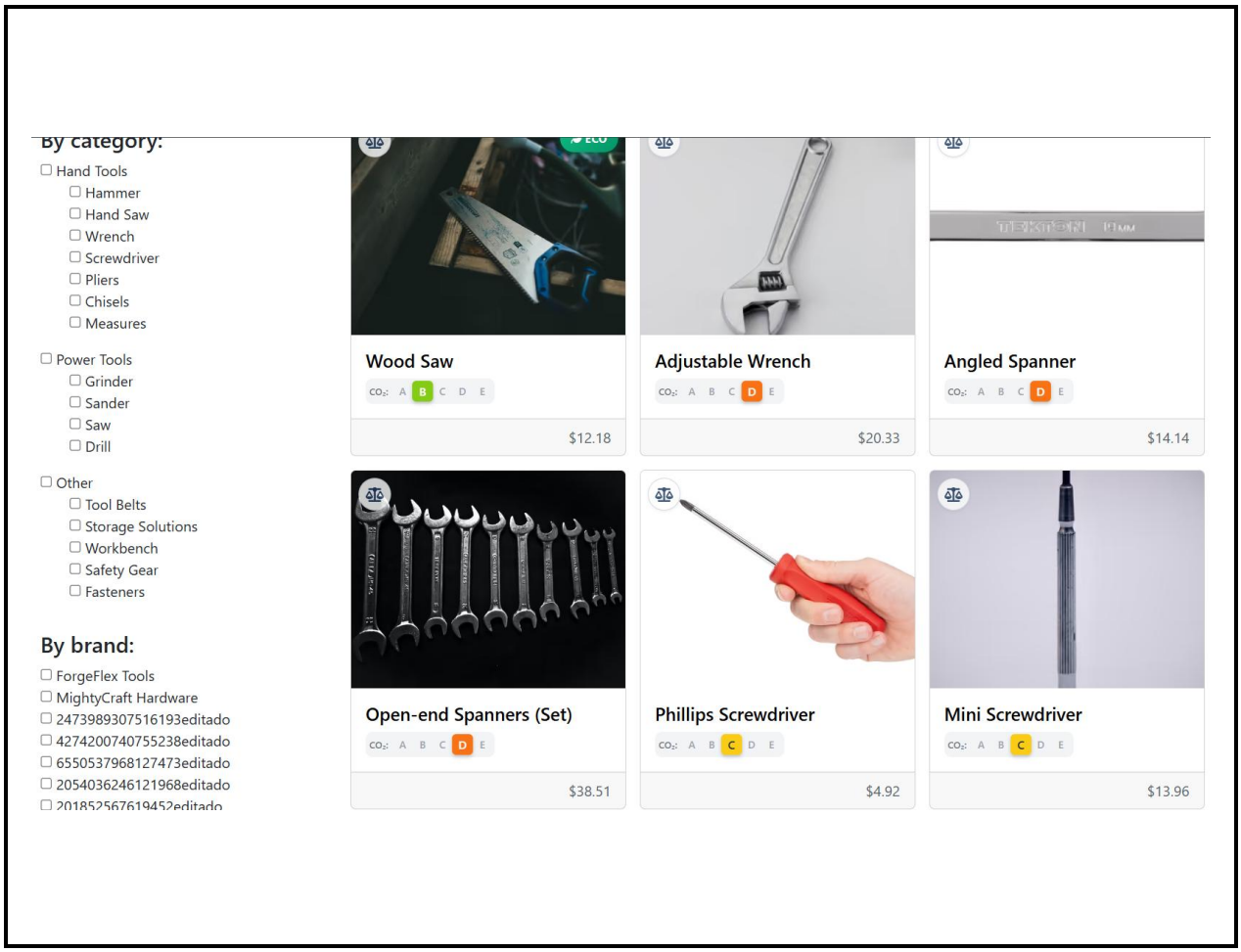


Figure Placeholder: 3. Product Details Specification View



Photo by Akshat Swaminath on Unsplash.

Open-end Spanners (Set)

Wrench | ForgeFlex Tools

$38.51

CO_2: A B C **D** E

Complete professional set of 12 open-end spanners covering metric sizes from 6mm to 32mm, crafted from chrome vanadium steel for lasting strength and resistance to flexing under heavy torque. Each spanner features a slim-profile head design angled at 15 degrees for access in tight spaces, and the open-end configuration allows quick lateral placement on fasteners without the need to thread them through from the end. A mirror-polished chrome finish resists corrosion, repels grease and oil, and looks sharp in any professional tool collection. Size markings are deeply laser-etched on both sides of each spanner for instant identification even in low light conditions. The set is supplied in a durable roll-up canvas pouch with individual pockets for organized storage and easy transport to any job site. An essential investment for automotive workshops, industrial maintenance departments, and serious home mechanics.

− 1 +

Add to cart | Add to favourites | Compare

Figure Placeholder: 4. Search and Multi-Faceted Filters

↕ Sort

Price (High - Low)

↔ Price Range

1 100 200

Search

Search X Search

Filters

By category:

- ☐ Hand Tools
  - ☑ Hammer
  - ☐ Hand Saw
  - ☐ Wrench
  - ☐ Screwdriver
  - ☐ Pliers
  - ☐ Chisels
  - ☐ Measures
- ☐ Power Tools

Searched for: hammer

7 products found for 'hammer'



Claw Hammer with Fiberglass Handle

CO_2: A B C **D** E

$20.14



Court Hammer

CO_2: A B C **D** E

$18.63



Sledgehammer

CO_2: A B C **D** E

$17.75







Figure Placeholder: 5. Shopping Cart Staging View

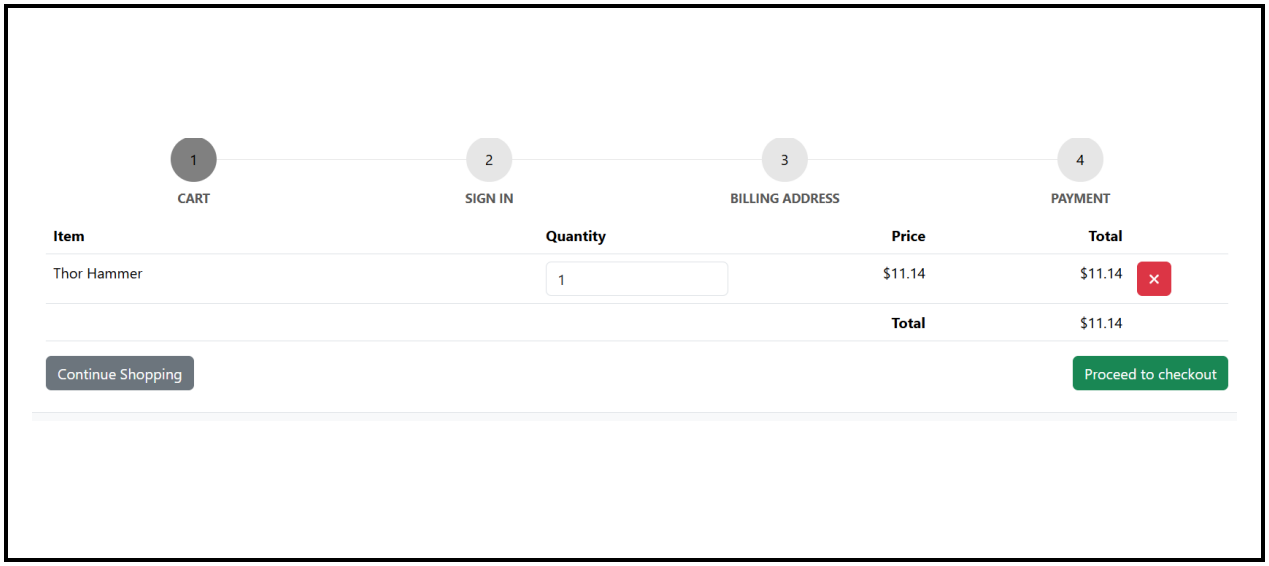


Figure Placeholder: 6. User Account (Sign-In / Registration Form)

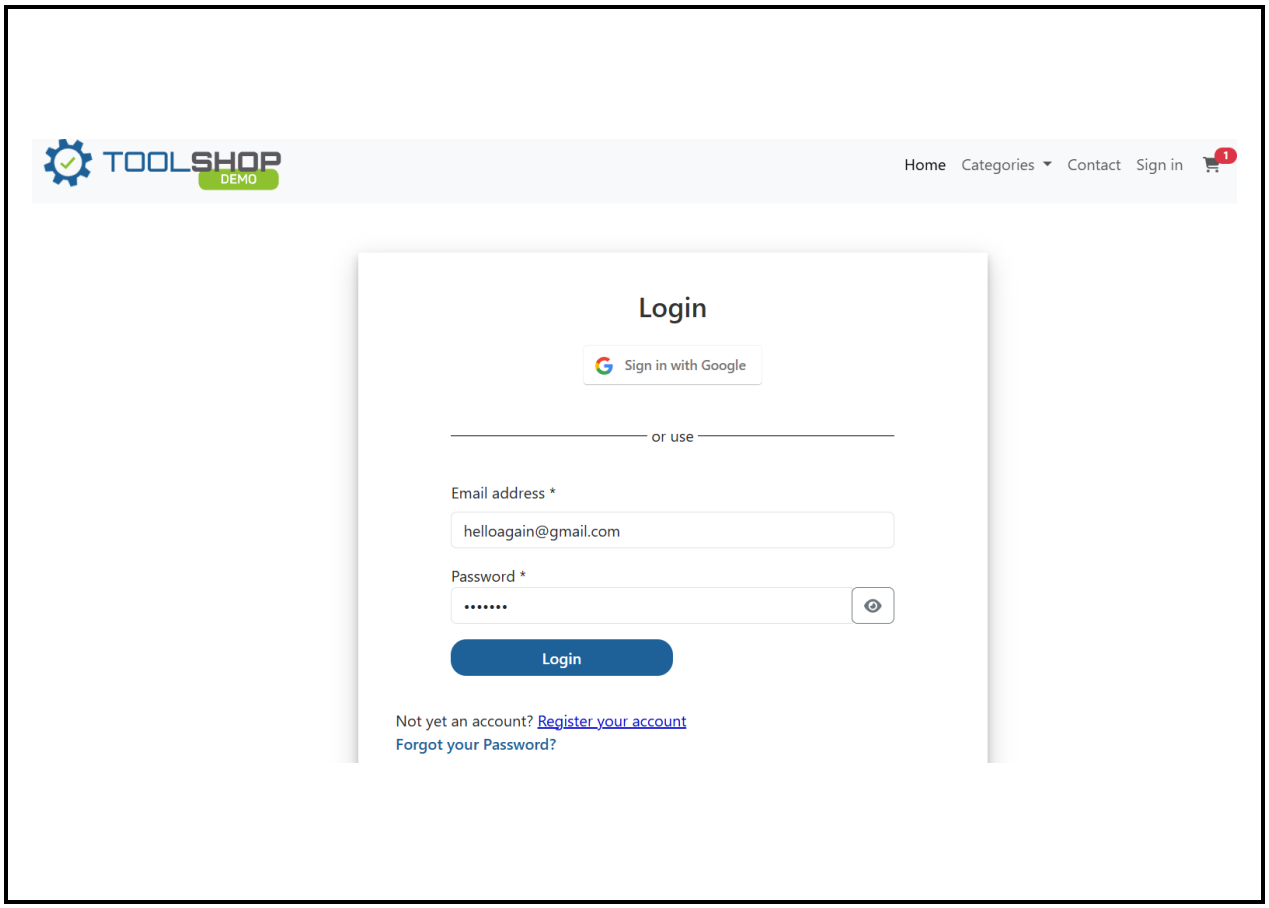


Figure Placeholder: 7. Checkout / Order Processing Funnel

